随机过程

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Finite-state Markov Chains

§ **Introduction**
§ Classification of States
§ The Matrix Representation

§ Introduction

思考题

设 $\mathcal{X}$ 是一个有限集，思考如下问题的描述复杂度：

1. 如何刻画 $\mathcal{X}$ 上的一个离散随机变量？
2. 如何刻画 $\mathcal{X}^n$ 上的离散随机向量？
3. 如何刻画 $\mathcal{X}^\infty$ 上的离散随机序列？
4. 如何刻画一个独立随机向量？
5. 如何刻画一个独立同分布的随机向量？

马尔可夫过程：

$$
\begin{aligned}
&P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) \\
&= P(X_1 = x_1) \cdot P(X_2 = x_2 | X_1 = x_1) \cdot P(X_3 = x_3 | X_1 = x_1, X_2 = x_2) \cdot \cdots \\
&= P(X_1 = x_1) \cdot P(X_2 = x_2 | X_1 = x_1) \cdot P(X_3 = x_3 | X_2 = x_2) \cdot \cdots
\end{aligned}
$$

§ Introduction

Definition 1

A Markov chain is an integer-time process, $\{X_n, n \geqslant 0\}$, for which the sample values for each rv. $X_n, n \geqslant 1$, lie in a countable set S and depend on the past only through the most recent rv. X_{n-1}. More specifically, for all positive integers n, and for all choices of $i, j, \ldots, \ell$ in S,

$$P\{X_n = j | X_{n-1} = i, X_{n-2} = k, \ldots, X_0 = \ell\} = P\{X_n = j | X_{n-1} = i\}$$

for all conditioning events $X_{n-1} = i, X_{n-2} = k, \ldots, X_0 = \ell$ of positive probability.

§ Introduction

Often we write the condition as follow

$$P\{X_n|X_{n-1},\ldots,X_0\} = P\{X_n|X_{n-1}\}.$$

The possible values of X_n are called states. The state at a given time summarizes everything about the past that is relevant to the future. Furthermore, $P\{X_n = j|X_{n-1} = i\}$ depends only on i and j (not n) and is denoted by

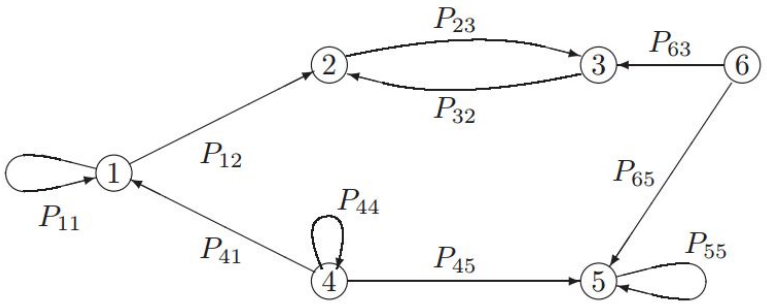
$$P\{X_n = j|X_{n-1} = i\} = P_{ij}.$$

Some people call this a homogeneous Markov chain.
For a non-homogeneous Markov chain, it satisfies

$$P\{X_n = j|X_{n-1} = i\} = P_{ij}(n).$$

§ Introduction

A Markov chain is completely described by the transition probabilities $\{P_{ij} : 1 \leqslant i, j \leqslant M\}$ plus the initial probabilities $P_{X_0}(i)$.
Sometimes we visualize $\{P_{ij}\}$ in terms of a directed graph and sometimes as a matrix.



$$[P] = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{16} \\ P_{21} & P_{22} & \cdots & P_{26} \\ \vdots & \vdots & \vdots\vdots\vdots & \vdots \\ P_{61} & P_{62} & \cdots & P_{66} \end{bmatrix}$$

Remark:

$$P_{ij} \geq 0 \quad \text{for} \quad 1 \leq i, j \leq M.$$

and

$$\sum_{\ell=1}^{M} P_{i\ell} = 1 \quad \text{for } 1 \leq i \leq M.$$

行和 $= 1$，列和未必。

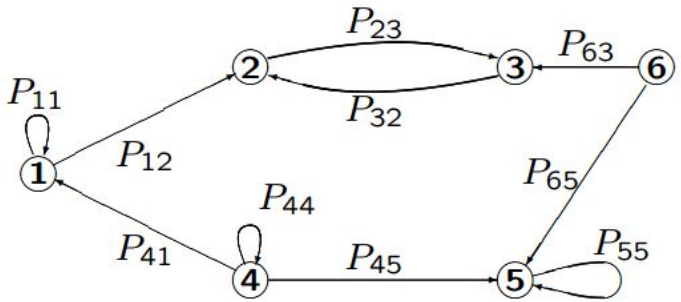
Finite-state Markov Chains

§ Introduction
§ **Classification of States**
§ The Matrix Representation

Classification of States

Definition 2

- An (n-step) walk is an ordered string of nodes, $(i_0, i_1, \ldots, i_n)$, $n \geqslant 1$, in which there is a directed arc from i_{m-1} to i_m for each $1 \leqslant m \leqslant n$.
- A path is a walk in which no nodes are repeated.
- A cycle is a walk in which the first and last nodes are the same and no other node is repeated.



Walk: (4, 4, 1, 2, 3, 2)
Walk: (4, 1, 2, 3)
Path: (4, 1, 2, 3)
Path: (6, 3, 2)
Cycle: (2, 3, 2)
Cycle: (5, 5)

Remark: an edge from i to j explicitly means that $P_{ij} > 0$.

§ Classification of States

Definition 3

A state j is accessible from i (abbreviated as $i \to j$) if there is a walk in the graph from i to j.

We denote $\Pr\{X_n = j | X_0 = i\}$ by P_{ij}^n. Then if i, k, j is a walk, $P_{ik} > 0$ and $P_{kj} > 0$,

$$P_{ij}^2 \geqslant P_{ik} P_{kj} > 0.$$

Similarly, if there is an n-step walk from i to j, then $P_{ij}^n > 0$.
Thus,

- $i \to j$: $P_{ij}^n > 0$ for some integer $n \geqslant 1$.
- $i \nrightarrow j$: $P_{ij}^n = 0$ for all integer $n \geqslant 1$.
- $P_{ik}^n > 0$ and $P_{kj}^m > 0$ imply $P_{ij}^{m+n} > 0$.
- $i \to k$ and $k \to j$ imply $i \to j$.

思考题： 给定概率转移矩阵，P^n 表示的意义是什么？如何计算 P^n?

§ Classification of States

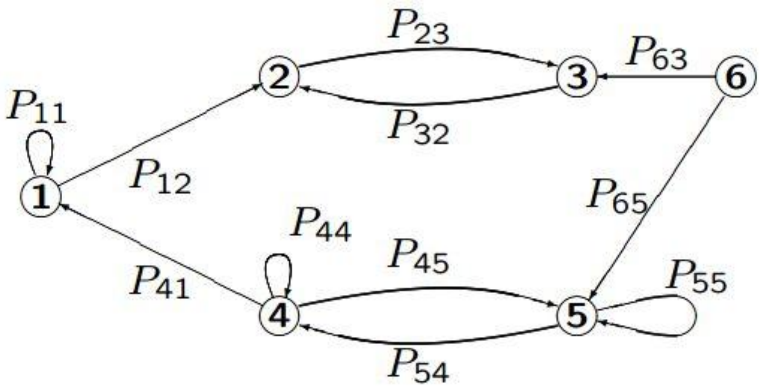
Definition 4

Two distinct states i and j communicate (abbreviated $i \leftrightarrow j$) if i is accessible from j and j is accessible from i.

- if $i \leftrightarrow j$, then $j \leftrightarrow i$.
- if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

Definition 5

A class C of states is a non-empty set of states such that for each $i \in C$, each state $j \neq i$ satisfies $j \in C$ if $i \leftrightarrow j$ and $j \notin C$ if $j \nleftrightarrow i$.

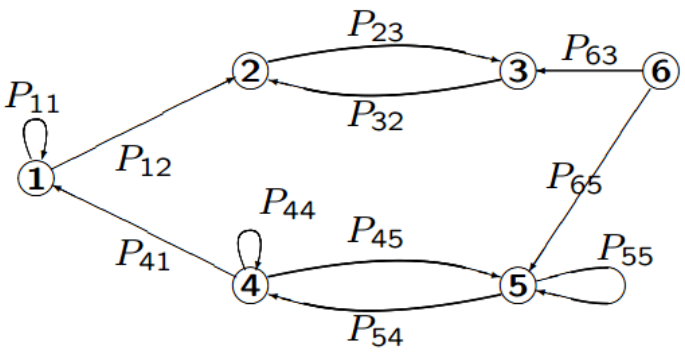


$\mathcal{C}_1 = \{2, 3\}$
$\mathcal{C}_2 = \{4, 5\}$
$\mathcal{C}_3 = \{1\}$
$\mathcal{C}_4 = \{6\}$

§ Classification of States

Definition 6

For finite-state Markov chains, a recurrent state is a state i that is accessible from all states that are accessible from i (i is recurrent if $i \to j$ implies that $j \to i$). A transient state is a state that is not recurrent.



2 and 3 are recurrent
4 and 5 are transient
$4 \to 1, 5 \to 1, 1 \not\to 4, 5$
6 and 1 also transient

Remark: If a Markov chain ever enters a recurrent state, it returns to that state eventually with probability 1, and thus keeps returning infinitely often. In fact, this essentially serves as the definition of recurrence for Markov chains with countably-infinite state spaces.

§ Classification of States

Theorem 1

For finite-state Markov chains, either all states in a class are transient or all are recurrent.

Proof:

- Assume that i is transient (i.e., for some j, $i \to j$ but $j \nrightarrow i$) and suppose that i and m are in the same class (i.e., $i \leftrightarrow m$).
- Then $m \to i$ and $i \to j$, so $m \to j$.
- If $j \to m$, then the walk from j to m could be extended to i; this is a contradiction, and therefore there is no walk from j to m, and m is transient.

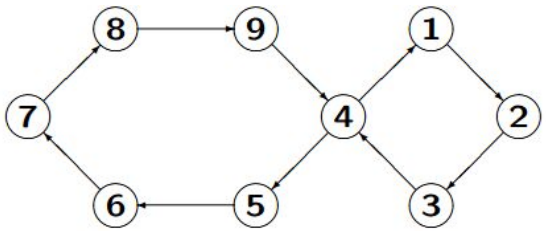
Periodic States and Classes

Definition 7

The period of a state i, denoted $d(i)$, is the greatest common divisor of those values of n for which $P_{ii}^n > 0$.

$$d(i) = \gcd\{n : P_{ii}^n > 0\}$$

If $d(i) = 1$, the state i is aperiodic. If $d(i) > 1$, i is periodic with period $d(i)$.



For example, $P_{44}^n > 0$ **for** n **= 4, 6, 8, 10;** $d(4) = 2$

For state 7, $P_{77}^n > 0$ **for** n **= 6, 10, 12, 14;** $d(7) = 2$

Periodic States and Classes

设 m, n 是互素的正整数，则存在正整数 T_0, 使所有的正整数 $T \geq T_0$ 都可以由 m, n 的非负组合表示，即，

$$\forall T \geq T_0,\ \exists u, v \in \mathbb{N},\ such\ that\ um + vn = T.$$

证明：
由 m, n 互素（不妨设 $m > n$），则存在正整数 s, t，使得 $sm - tn = 1$.
令 $T_0 = tn^2$，则对于所有 $T \geq T_0$，由欧几里得算法，得

$$T = qn + r, q \geq tn, 0 \leq r < n.$$

所以

$$T = qn + r(sm - tn) = (q - rt) \cdot n + rs \cdot m$$

Periodic States and Classes

Theorem 2

For any Markov chain (with either a finite or countably-infinite number of states), all states in the same class have the same period.

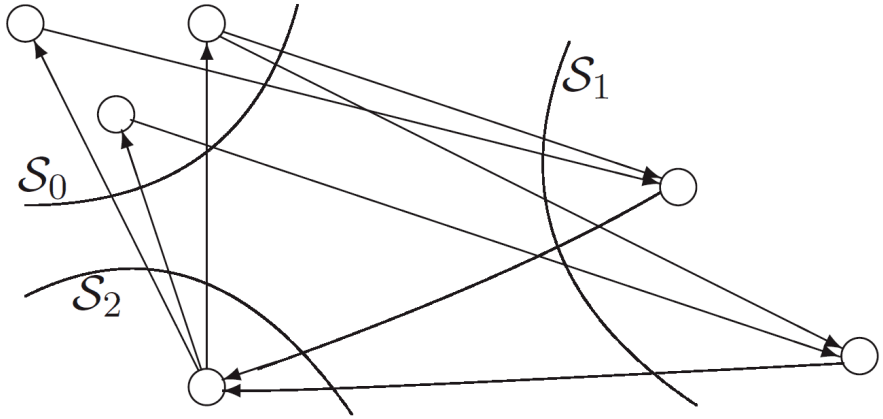
Proof: Let i and j be any distinct pair of states in a class C. Then $i \leftrightarrow j$ and there is some n such that $P_{ij}^n > 0$ and some m such that $P_{ji}^m > 0$.

- Since there is a walk of length $n+m$ going from i to j and back to i, $n+m$ must be divisible by $d(i)$.
- Let t be any integer such that $P_{jj}^t > 0$. Since there is a walk of length $n+t+m$ from i to j, then back to j, and then to i, $n+t+m$ is divisible by $d(i)$, and thus t is divisible by $d(i)$.
- Since this is true for every t such that $P_{jj}^t > 0$, $d(j)$ is divisible by $d(i)$.
- Reversing the roles of i and j, $d(i)$ is divisible by $d(j)$, so $d(i) = d(j)$.

Periodic States and Classes

Theorem 3

If a class C in a finite-state Markov chain has period d, then the states in C can be partitioned into d subsets, $S_0, S_1, \ldots, S_{d-1}$, in such a way that all transitions from S_ℓ go to $S_{\ell+1}$ for $\ell < d-1$, and go to S_0 for $\ell = d-1$.



In other words, starting in a given subclass, the state cycles through the d subclasses.

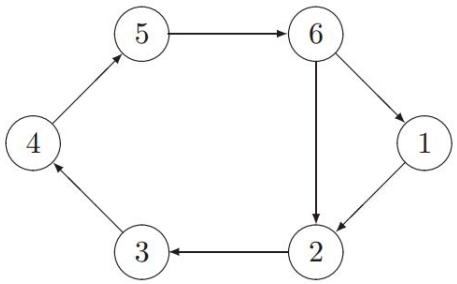
Ergodic Class of States

Definition 8

For a finite-state Markov chain, an ergodic class of states is a class that is both recurrent and aperiodic. A Markov chain consisting entirely of one ergodic class is called an ergodic chain.

Theorem 4

For an ergodic M state Markov chain, $P_{ij}^m > 0$ for all i, j, and all $m \geqslant (M-1)^2 + 1$.



An ergodic chain with $M = 6$ states in which $P_{ij}^m > 0$ for all $m > (M-1)^2$ and all i, j but $P_{11}^{(M-1)^2} = 0$.

Ergodic Class of States

Theorem 4

For an ergodic M state Markov chain, $P_{ij}^m > 0$ for all i, j, and all $m \geqslant (M-1)^2 + 1$.

Proof: a) Show that an ergodic Markov chain with $M > 1$ states must contain a cycle with $\tau < M$ states.

- A recurrent chain must contain cycles.
- The states in any cycle (not counting the initial state) are distinct and thus the number of steps in a cycle is at most M.
- If the chain only contained cycles with M states, then it would be periodic with period M.

Ergodic Class of States

b) Let ℓ be a fixed state on a fixed cycle of length $\tau < M$. Let $\mathcal{T}(m)$ be the set of states accessible from ℓ in m steps. Show that for each $m \geqslant 1$, $\mathcal{T}(m) \subseteq \mathcal{T}(m+\tau)$.

- For any state $j \in \mathcal{T}(m)$, we have

$$P_{\ell j}^{m} > 0.$$

Therefore,

$$P_{\ell j}^{m+\tau} \geqslant P_{\ell\ell}^{\tau} P_{\ell j}^{m} > 0.$$

c) Define $\mathcal{T}(0)$ to be the singleton set $\{\ell\}$ and show that

$$\mathcal{T}(0) \subseteq \mathcal{T}(\tau) \subseteq \mathcal{T}(2\tau) \subseteq \cdots \subseteq \mathcal{T}(n\tau) \subseteq \ldots.$$

Ergodic Class of States

d) Show that if one of the inclusions above is satisfied with equality, then all subsequent inclusions are satisfied with equality. Show from this that at most the first $M-1$ inclusions can be satisfied with strict inequality and that $\mathcal{T}(n\tau) = \mathcal{T}((M-1)\tau)$ for all $n \geqslant M-1$.

- It can be proved by induction that if $\mathcal{T}((k+1)\tau) = \mathcal{T}(k\tau)$ for some k, then $\mathcal{T}(n\tau) = \mathcal{T}(k\tau)$ for $n \geqslant k$.
- Let k be the smallest integer for which $\mathcal{T}((k+1)\tau) = \mathcal{T}(k\tau)$. For $n \leqslant k$, the size of $\mathcal{T}(n\tau)$ must increase as n increases. Since $|\mathcal{T}(0)| = 1$, we have $k+1 \leqslant |\mathcal{T}(k\tau)| \leqslant M$.

Ergodic Class of States

e) Show that all states are included in $T((M-1)\tau)$.

- Since ℓ is aperiodic, $\exists \lambda > 0$, such that

$$P_{\ell\ell}^{\lambda} > 0 \quad \text{and} \quad a\lambda = b\tau + 1$$

  for some positive integer a, b.

- For any state j, $\exists k > 0, P_{\ell j}^{k} > 0$. Let

$$k' = k \mod \tau.$$

  Then we have

$$P_{\ell j}^{(\tau-k')a\lambda+k} \geqslant P_{\ell\ell}^{(\tau-k')a\lambda} P_{\ell j}^{k} > 0,$$

  where $\tau | [(\tau - k')a\lambda + k]$, so $j \in T((M-1)\tau)$.

Ergodic Class of States

f) Show that $P_{ij}^{(M-1)^2+1} > 0$ for all i, j.

- For the state ℓ contained in a cycle with $\tau \leqslant M-1$ states:
  - Let ℓ' be the state that (ℓ, ℓ') is a transition on this cycle, then $\mathcal{T}((M-1)\tau+1) \supseteq \mathcal{T}'((M-1)\tau) = \mathcal{S}$.
  - It can be proved by induction $\mathcal{T}(m) = \mathcal{S}$, for $m \geqslant (M-1)\tau$.
  - $(M-1)^2+1 > (M-1)\tau$, so $P_{\ell j}^{(M-1)^2+1} > 0$.
- For the state ℓ^* not contained in that cycle, it can reach the cycle at most in $(M-\tau)$ steps. We have

$$\mathcal{T}^*((M-1)^2+1) \supseteq T((M-1)^2+1-(M-\tau)).$$

Since $(M-1)^2+1-(M-\tau)-(M-1)\tau = M(M-1-\tau) \geqslant 0$, we have $\mathcal{T}^*((M-1)^2+1) = \mathcal{S}$.

Finite-state Markov Chains

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§ The Matrix Representation

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§ Steady state assuming $[P] > 0$
§ Ergodic Markov chains
§ Ergodic unichains
§ Arbitrary finite-state Markov chains

§ The Matrix Representation

- The matrix $[P]$ of transition probabilities of a Markov chain is called a stochastic matrix.

$$\sum_{\ell=1}^{M} P_{i\ell} = 1 \quad \text{for } i = 1, 2, \ldots, M.$$

- For 2-step walks, using the Markov property,

$$P_{ij}^2 = \sum_{k=1}^{M} P_{ik} P_{kj}.$$

The matrix of 2-step transition probabilities $[P^2] = [P][P]$.

- The Chapman-Kolmogorov equation:

$$[P^{m+n}] = [P^m][P^n]$$

$$P_{ij}^{m+n} = \sum_{k=1}^{M} P_{ik}^m P_{kj}^n.$$

Steady state and $[P^n]$ for large n

If memory of the past dies out with increasing n, then we would expect the dependence of P_{ij}^n on both n and i to disappear as $n \to \infty$,

$$\lim_{n\to\infty} P_{ij}^n = \pi_j \quad \text{for } 1 \leqslant i, j \leqslant M,$$

where π_j is a function only of j and not of i or n.
we can multiply both sides by P_{jk} and sum over j, getting

$$\lim_{n\to\infty} \sum_j P_{ij}^n P_{jk} = \sum_j \pi_j P_{jk}.$$

The left side is $\lim_{n\to\infty} P_{ik}^{n+1} = \pi_k$. Thus if this limit exists, the vector $\boldsymbol{\pi}$ must satisfy

$$\pi_k = \sum_j \pi_j P_{jk} \quad \text{for each } k.$$

Steady state and $[P^n]$ for large n

Definition 9

A steady-state vector (or a steady-state distribution) for an M-state Markov chain with transition matrix $[P]$ is a row vector $\boldsymbol{\pi}$ that satisfies

$$\boldsymbol{\pi} = \boldsymbol{\pi}[P],$$

where $\sum_i \pi_i = 1$ and $\pi_i \geqslant 0$ for $1 \leqslant i \leqslant M$.

Remark: It is important to recognize that if $\boldsymbol{\pi}$ is a steady-state vector, it does NOT necessarily follow that $[P^n]$ converges to $\boldsymbol{\pi}$ as $n \to \infty$.

- Under what conditions does $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ have a probability vector solution?
- Under what conditions does $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ have a unique probability vector solution?
- Under what conditions does each row of $[P^n]$ converge to a probability vector $\boldsymbol{\pi}$ satisfying $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$?

Steady state and $[P^n]$ for large n

Definition 10

A unichain is a finite-state Markov chain that contains a single recurrent class plus, perhaps, some transient states.
An ergodic unichain is a unichain for which the recurrent class is ergodic.

- First, $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ always has a probability vector solution (although this is NOT necessarily true for infinite-state chains).
- Second, $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ has a unique probability vector solution iff $[P]$ is the transition matrix of a unichain.
- Third, each row of $[P^n]$ converge to the unique solution of $\boldsymbol{\pi} = \boldsymbol{\pi}[P]$ iff $[P]$ is the transition matrix of an ergodic unichain.

Steady state and $[P^n]$ for large n

Lemma 1

Let $[P]$ be the transition matrix of a finite-state Markov chain and let $[P^n]$ be the nth power of $[P]$. Then for each state j and each integer $n \geqslant 1$,

$$\max_i P_{ij}^{n+1} \leqslant \max_\ell P_{\ell j}^n, \qquad \min_i P_{ij}^{n+1} \geqslant \min_\ell P_{\ell j}^n.$$

Proof: Use the Chapman–Kolmogorov equation, for any state i, j and any step n, we have

$$P_{ij}^{n+1} = \sum_k P_{ik} P_{kj}^n \leqslant \sum_k P_{ik} \max_\ell P_{\ell j}^n = \max_\ell P_{\ell j}^n.$$

Maximizing i, so

$$\max_i P_{ij}^{n+1} \leqslant \max_\ell P_{\ell j}^n.$$

Replacing maxima with minima and reversing inequalities,

$$\min_i P_{ij}^{n+1} \geqslant \min_\ell P_{\ell j}^n.$$

Steady state assuming $[P] > 0$

Lemma 2

Let $[P] > 0$ be the transition matrix of a finite-state Markov chain and let $\alpha = \min_{i,j} P_{ij}$. Then for all states j and all $n \geqslant 1$:

$$\max_i P_{ij}^{n+1} - \min_i P_{ij}^{n+1} \leqslant \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n\right)(1-2\alpha).$$

$$\left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n\right) \leqslant (1-2\alpha)^n.$$

$$\lim_{n\to\infty} \max_\ell P_{\ell j}^n = \lim_{n\to\infty} \min_\ell P_{\ell j}^n \geqslant \alpha > 0.$$

Steady state assuming $[P] > 0$

Proof: We tighten the proof of Lemma 1 slightly. For a given n and j, let $\ell_{\min}$ be a state that minimizes P_{ij}^n over i.

$$
\begin{aligned}
P_{ij}^{n+1} &= \sum_k P_{ik} P_{kj}^n \\
&\leqslant \sum_{k \neq \ell_{\min}} P_{ik} \max_\ell P_{\ell j}^n + P_{i\ell_{\min}} \min_\ell P_{\ell j}^n \\
&= (1 - P_{i\ell_{\min}}) \max_\ell P_{\ell j}^n + P_{i\ell_{\min}} \min_\ell P_{\ell j}^n \\
&= \max_\ell P_{\ell j}^n - P_{i\ell_{\min}} \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \right) \\
&\leqslant \max_\ell P_{\ell j}^n - \alpha \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \right). \\
\max_i P_{ij}^{n+1} &\leqslant \max_\ell P_{\ell j}^n - \alpha \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \right).
\end{aligned}
$$

Steady state assuming $[P] > 0$

We have shown that

$$\max_i P_{ij}^{n+1} \leqslant \max_\ell P_{\ell j}^n - \alpha \left(\max_\ell P_{\ell j}^n - \min_l P_{\ell j}^n \right).$$

Interchanging max with min and $\leqslant$ with $\geqslant$, we get

$$\min_i P_{ij}^{n+1} \geqslant \min_\ell P_{\ell j}^n + \alpha \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \right).$$

Subtracting these equations,

$$\max_i P_{ij}^{n+1} - \min_i P_{ij}^{n+1} \leqslant \left(\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \right) (1 - 2\alpha).$$

Since $\min_l P_{lj} \geqslant \alpha$ and $\max_l P_{lj} \leqslant 1 - \alpha$,

$$\max_i P_{ij} - \min_i P_{ij} \leqslant 1 - 2\alpha.$$

Iterating on n,

$$\max_\ell P_{\ell j}^n - \min_\ell P_{\ell j}^n \leqslant (1 - 2\alpha)^n.$$

Ergodic Markov chains

Theorem 5

Let $[P]$ be the matrix of an ergodic finite-state Markov chain. Then there is a unique steady-state vector $\boldsymbol{\pi}$ satisfies

$$\lim_{n\to\infty} P_{ij}^n = \pi_j \quad \text{for each } i, j,$$

with geometric convergence in n.

The key to this theorem is the pair of statements that $\max_i P_{ij}^n$ is non-increasing in n and $\min_i P_{ij}^n$ is non-decreasing in n.

Ergodic Markov chains

Proof:

We have shown that $[P^h]$ is positive for $h = (M-1)^2 + 1$, so we can apply Lemma 2 to $[P^h]$, with $\alpha = \min_{ij} P^h_{ij}$.

We don't much care about the value of α, but only that it is positive. Then

$$\lim_{m\to\infty} \max_{\ell} P^{hm}_{\ell j} = \lim_{m\to\infty} \min_{\ell} P^{hm}_{\ell j} \triangleq \pi_l \geqslant \alpha > 0.$$

To show that the limit applies for all n rather than than just multiples of h, we use Lemma 1, showing that $\max_\ell P^n_{\ell j}$ is non-increasing in n, so it must have the same limit as $\max_\ell P^{hm}_{\ell j}$. The same argument applies for the minima.

Ergodic Markov chains

Then we need to show that $\boldsymbol{\pi}$ is the unique steady-state vector. Since each row of $[P^n]$ converges to this same vector $\boldsymbol{\pi}$. This is represented most compactly by

$$\lim_{n\to\infty}[P^n] = \boldsymbol{e}\boldsymbol{\pi}$$

where $\boldsymbol{e} = (1, 1, \ldots, 1)^T$.
Let $\boldsymbol{\mu}$ be any steady-state vector, i.e., any probability vector solution to $\boldsymbol{\mu}[P] = \boldsymbol{\mu}$. Then

$$\boldsymbol{\mu} = \boldsymbol{\mu}\lim_{n\to\infty}[P^n] = \boldsymbol{\mu}\boldsymbol{e}\boldsymbol{\pi} = \boldsymbol{\pi}.$$

using the fact that $\boldsymbol{\mu}\boldsymbol{e} = 1$.

Ergodic Unichains

Let T denote the set of transient states (which might contain several transient classes), and assume the states of T are numbered $1, 2, \ldots, t$. Let R denote the recurrent class, assumed to be numbered $t+1, \ldots, t+r$.

$$[P] = \begin{bmatrix} [P_T] & [P_{TR}] \\ [0] & [P_R] \end{bmatrix}.$$

The block of zeroes in the lower left corresponds to the absence of transitions from recurrent to transient states.

We will show that P_{ij}^n converges to 0 for $i, j \in T$.

Ergodic Unichains

For each transient state, there must be a walk to some recurrent state and since there are only t transient states, there must be some walk of length at most t. For each $i \in T$, $\sum_{j \in R} P_{ij}^t > 0$, i.e. $i \in T$,$\sum_{j \in T} P_{ij}^t < 1$. Thus,

$$\gamma \triangleq \max_{i \in T} \sum_{j \in T} P_{ij}^t < 1.$$

Lemma 3

Let $[P]$ be a unichain, T be the set of transient states, and let $t = |T|$.

$$\max_{\ell \in T} \sum_{j \in T} P_{\ell j}^n \leqslant \gamma^{\lfloor n/t \rfloor}.$$

Ergodic Unichains

Theorem 6

Let $[P]$ be the transition matrix of a finite-state ergodic unichain. Then $\lim_{n\to\infty}[P^n] = e\boldsymbol{\pi}$, where $e = (1, 1, \ldots, 1)^T$ and $\boldsymbol{\pi}$ is the steady-state vector of the recurrent class of states, expanded by zeros for each transient state of the unichain. The convergence is geometric in n for all i, j.

Arbitrary Finite-state Markov Chains

- First consider a Markov chain with several recurrent classes, $R_1, \ldots, R_m$. The classes don' t communicate and should be considered separately.
- If one insists on analyzing the entire chain, $[P]$ will have m independent steady state vectors, one nonzero on each class.
- Next consider a periodic recurrent chain of period d. This can be separated into d subclasses with a cyclic rotation between them.
- If we look at $[P^d]$, we see that each subclass becomes an ergodic class, say $C_1, \ldots, C_d$. Thus $\lim_{n\to\infty}[P^{dn}]$ exists.

课堂练习

1. 假设 $\mathbf{X} = (X_n, n \geq 0)$ 是一个状态空间为 $\{1, 2, 3\}$ 的 Markov 链，转移矩阵为

$$\mathbf{P} = \begin{pmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.1 & 0.4 \\ 0.5 & 0.2 & 0.3 \end{pmatrix}$$

如果 $\boldsymbol{p}_0 = (0.5, 0.5, 0)$，求

（1）$P(X_0 = 2, X_1 = 2, X_2 = 1)$.

（2）$P(X_1 = 2, X_2 = 2, X_3 = 1)$.

课堂练习

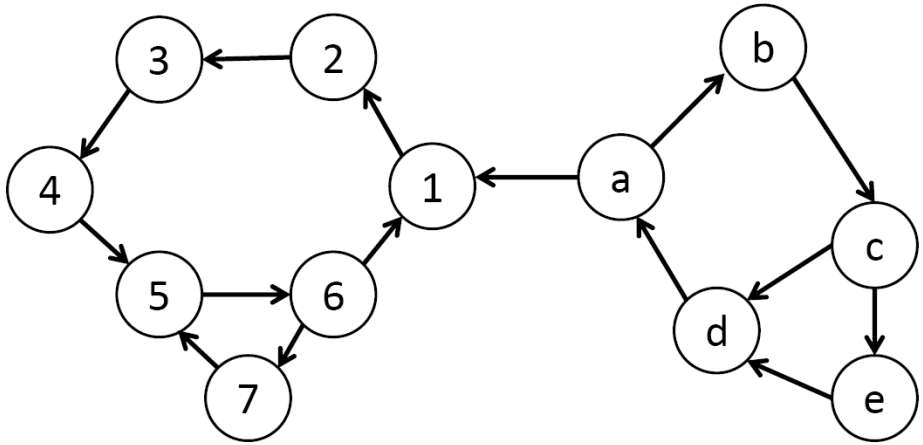
2. 假设 $\mathbf{X} = (X_n, n \geq 0)$ 是 Markov 链，状态空间为 $\{1, 2, 3, 4\}$，转移概率矩阵为

$$\mathbf{P} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}$$

问：哪些状态是常返的，哪些是瞬时的？并求每个状态的周期。

课堂练习

3. 考虑下图马尔可夫链，问：



（1）状态可以分为几类？
（2）各类的周期分别是什么？
（3）哪些状态是常返的，哪些状态是瞬时的？

课堂练习

4. 假设 $\mathbf{X} = (X_n, n \geq 0)$ 是 Markov 链，请分别考虑以下两个转移矩阵

$$\mathbf{P1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \mathbf{P2} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

（1）分别对上述两个状态空间进行分类；
（2）分别列出上述两个状态空间各自对应的连通类；
（3）上述两个状态空间各自连通类之间互通吗？

课堂练习

5. 某地区的天气状况可分为多云和阴天两种，分别记为 1 和 2，并用 Markov 链 $\mathbf{X} = (X_n, n \geq 0)$ 描述，其转移概率矩阵为

$$\mathbf{P} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

如果已知今天是多云，求 3 天、5 天、10 天及更长时间以后天气状况的分布规律。

Thank you for your attention!